

Factors Affecting Photosynthesis

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation to save screenshots of the tank, charts or measurement series and insert them in the marked spaces.

Use the simulation to investigate how light, CO₂ supply and temperature affect photosynthesis. Bacteria gather where the alga produces most oxygen. Use the magnifier, charts and protocol, and change only one factor at a time.

Task block 1: Influence of light

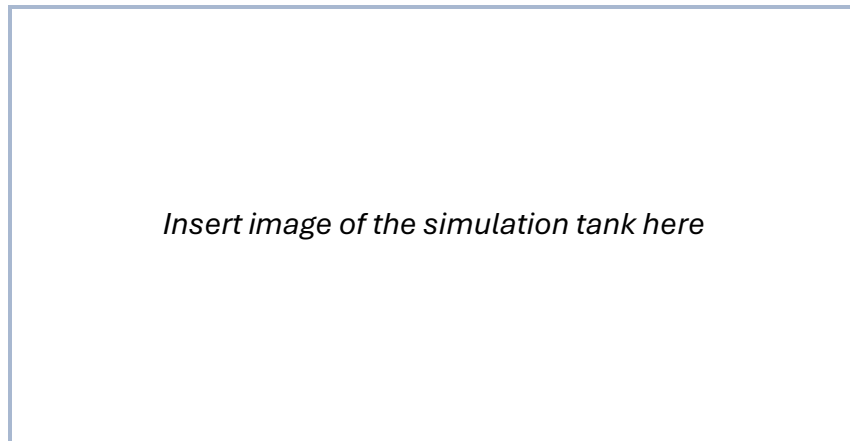
a) Light colours and prism

- Set the light colour to white and activate the prism.
- Choose a high light intensity so that differences are clearly visible.
- Move the magnifier along the spectrum and observe relative photosynthesis and bacterial accumulation.
- Record the colour or wavelength ranges with especially high or low photosynthetic activity.

Area / colour studied	Photosynthetic activity	Observation
1.		
2.		
3.		
Wavelength	Maximum bei: Minimum bei:	

Task: Which light ranges promote photosynthesis most strongly?

Wavelength with maximum photosynthesis rate (simulation tank)



b) Light intensity

- Set the temperature slider to 35 deg C and the CO₂ supply slider to 100%.
- Work with white light and without the prism.
- Switch to "apparent photosynthesis".
- Now change only the light intensity according to the table.

Task: Explain the difference between real and apparent photosynthesis.

Beleuchtungs- stärke	Photosynthetic activity
20 %	
40%	
60%	
100%	

Task: Explain compensation point and saturation point.

Light-intensity chart at 100%

Insert here

Task block 2: Influence of temperature

2a) Set the temperature to the optimum

- Set light intensity and CO₂ supply to 100% and activate "apparent photosynthesis".
- Work with white light (with prism) and place the magnifier on the optimum wavelength (Task 1a).
- Investigate temperature dependence and determine the range with the highest activity.

Temperature	Photosynthetic activity
0 °C	
10 °C	
20 °C	
30 °C	
40 °C	
50 °C	

Task: Determine the temperature optimum and justify your decision.

b) Light intensity and temperature

Set the temperature to the optimum from Task 2a, activate "real photosynthesis" and keep the other settings.

- Now change only the light intensity according to the table and describe the temperature chart.

Beleuchtungsstärke	Photosynthetic activity	Description of the curve
20 %		
40 %		
60 %		
80 %		

Task: How do the temperature curves differ at 40% and 80% light intensity?

Temperature chart at a light intensity of 40%

Insert here

Temperature chart at a light intensity of 80%

Insert here

Task: Form a hypothesis for the different curve shapes at 40% and 80%.

Task block 3: CO₂ supply

- Set light to maximum and temperature to the optimum. Use white light without prism.
- Switch to "apparent photosynthesis".
- Vary CO₂ supply systematically.

CO2 supply	Photosynthetic activity
0 %	
25%	
50%	
75%	
100%	

Task: Determine the CO₂ compensation point and explain apparent photosynthesis = 0%.

Screenshot 3: CO₂ chart at 100% CO₂ supply

Insert here